

PRODUCT REQUIREMENTS DOCUMENT

LeadForge

Local Small Business Lead Generation, Scoring, Acquisition & Outreach Pipeline

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1. Product Overview

1.1 Problem Statement

Hyper-local small businesses in underserved commercial niches — septic services, tire shops, barbershops, towing companies, nail salons, and similar trades — represent a significant but underexploited market for digital marketing services. These businesses share three structural characteristics that make them simultaneously high-value targets and difficult to reach via conventional channels:

- They operate with minimal or nonexistent digital infrastructure, meaning standard inbound marketing funnels do not reach them.
- They are typically owner-operated with constrained time and attention, making cold outreach the only viable acquisition channel.
- Their revenue models are geographically bound, which means marketing ROI is unusually legible — if you can demonstrate local visibility, the value proposition is self-evident.

The challenge is systematic identification, qualification, prioritization, and automated initial contact of these businesses at a granular geographic level (Chicago zip codes) with enough signal fidelity to predict acquisition likelihood and appropriate price points before committing human time.

1.2 Product Definition

LeadForge is an internal application that automates the full lifecycle from lead identification to first contact. It uses LLM-powered scraping to extract and normalize business data from public sources, applies a composite scoring model to predict acquisition probability and price sensitivity, deploys an AI voice agent to initiate cold outreach and schedule meetings, captures conversation sentiment to refine scoring, and manages the full pipeline through an integrated lightweight CRM. Leads are organized by Chicago zip code and niche vertical.

1.3 Target Verticals

Vertical	License Type	Revenue Proxy	Digital Maturity
Septic tank services	IL Plumbing License	Service call volume	Very low
Used auto parts	Chicago Business License	Inventory turnover	Low
Small meat markets	IL Food License	Foot traffic / reviews	Low
Local bars	IL Liquor License	Review volume / events	Medium-low
Nail salons	IL Cosmetology License	Review volume / photos	Medium
Beauty shops	IL Cosmetology License	Review volume / social	Medium
Smoke shops	Chicago Tobacco License	Foot traffic	Low
Beauty supply stores	Chicago Business License	Foot traffic	Low
Mobile mechanics	None required (IL)	Service area radius	Very low
Tire shops	Chicago Business License	Review volume	Low

Lawn services	Pesticide Applicator (if appl.)	Seasonal demand	Low
Towing companies	IL Relocation Towing License	Police/AAA contracts	Low
Barbershops	IL Barber License	Review volume / chair count	Medium
Veterinarians	IL Veterinary License	Review volume / specialization	Medium
Security services	IL PERC Card / Agency License	Contract size	Low-medium

2. Data Architecture

2.1 Primary Data Sources

All data collection is restricted to publicly accessible sources. No purchased data, no scraping behind authentication walls (except where session cookies are used for public-facing content), no violation of platform terms that explicitly prohibit scraping.

Source	Data Available	Access Method	Status
Chicago Data Portal (data.cityofchicago.org)	Business licenses, food inspections, building permits, tobacco/liquor licenses	Socrata REST API	Verified ✓ Free, no auth
Google Places API	Business name, address, phone, website, hours, reviews, rating, photos, place ID	REST API	Verified ✓ Free tier 1,000/day
Illinois DFPR License Lookup	Professional licenses (cosmetology, barber, veterinary, plumbing)	Web form — Scraping headless automation	Verified ✓ Public, no API
Illinois Secretary of State	Business registration, incorporation date, registered agent, status	Web form — Scraping headless automation	Verified ✓ Public, no API
Yelp Fusion API	Reviews, ratings, categories, price range, transaction types	REST API	Verified ✓ Free tier 5,000/day
US Census / ACS	Zip demographics: median income, population density, age distribution, housing	Census Data API	Verified ✓ Free, unlimited
PageSpeed Insights API	Website quality score, mobile usability, load time	REST API	Verified ✓ Free
WHOIS / DNS lookup	Domain registration date, registrar, SSL status	Various APIs	Verified ✓ Free
Google Ads Transparency Center	Active Google Ads by advertiser	Public search	Verified ✓ Free

2.2 Supplemental Sources

Supplemental platform scraping defaults to custom Scraping scrapers to minimize per-unit cost. Apify actors are reserved for Meta products (Instagram, Facebook, Threads) where anti-bot countermeasures make custom scrapers impractical to maintain. Nextdoor cookie refresh is handled by an automated rotation service (see Section 4.3).

Source	Scraping Method	Data Available	Cost
Instagram	Apify: apify/instagram-scraper + Profile Scraper + Location Scraper + Hashtag Scraper	Profile existence, follower count, post frequency, bio, business status, posts tagged at business location, posts using business name as hashtag	~\$2.30/1,000 items
Facebook Pages	Apify: Facebook Posts Scraper + Facebook Search Scraper	Page existence, post frequency, engagement metrics, last post date,	Pay-per-compute

		customer check-ins, posts tagging business page, location-tagged UGC	
Threads	Apify: Meta Ads Library Scraper	Active ads across FB/IG/Threads/WhatsApp	Pay-per-compute
Meta Ad Library	Apify: Facebook Ads Library Scraper	All active ads by business name across Meta platforms	Pay-per-compute
Nextdoor	Scrapling StealthyFetcher + automated cookie refresh	Business ratings, recommendations, neighborhood, profile URLs	Free (infrastructure only)
Craigslist (Chicago services)	Scrapling Fetcher	Service listings, contact info, categories	Free (infrastructure only)
Thumbtack	Scrapling StealthyFetcher	Business name, location, rating, hires count, credentials	Free (infrastructure only)
Angi / Angie's List	Scrapling StealthyFetcher	Company details, ratings, reviews, contact info	Free (infrastructure only)

Supplemental Source Rationale

The primary sources (Chicago Data Portal + Google Places + Yelp) cover approximately 70–80% of businesses with any digital presence. The supplemental sources serve two distinct functions:

- Gap filling for low-digital-maturity niches: Mobile mechanics, lawn services, and independent towing companies frequently advertise on Craigslist and Thumbtack rather than maintaining Google Business Profiles. These platforms are the only discovery mechanism for the hardest-to-find leads.
- Enrichment for scoring precision: Nextdoor recommendations provide a hyperlocal trust signal that Google reviews do not capture. Angi/Thumbtack hire counts serve as direct revenue proxies. Instagram/Facebook activity data directly feeds the digital deficit score.
- Cost rationale: Custom Scrapling scrapers for Craigslist, Thumbtack, Angi, and Nextdoor eliminate per-item Apify charges entirely. At projected volumes of 2,000–5,000 leads/month, this saves an estimated \$150–400/month versus routing everything through Apify. Meta platforms remain on Apify because Instagram and Facebook deploy sophisticated bot detection that would require continuous scraper maintenance exceeding the Apify cost.

2.3 Data Model

The core data model consists of five primary entities: Business, Digital Presence, Competitive Context, Lead Score, and Outreach Record.

Business Entity

Field	Type	Source
business_id	UUID	Generated
name	String	Google Places / Chicago Data Portal (LLM-normalized)
address	String	Google Places
zip_code	String	Google Places / Chicago Data Portal
phone	String	Google Places / Scrapling enrichment
email	String	Website scrape / LLM extraction / Scrapling enrichment

owner_name	String	IL SOS / DFPR / website / LLM extraction
niche	Enum	Classified by license type + Google category
license_number	String	Chicago Data Portal / DFPR
license_status	Enum	Chicago Data Portal / DFPR (Active/Expired/Revoked/Unknown)
license_issue_date	Date	Chicago Data Portal / DFPR
incorporation_date	Date	IL Secretary of State
employee_count_est	Integer	LinkedIn / LLM estimate
estimated_monthly_revenue	Float	Derived (see Section 3.3)
google_place_id	String	Google Places API
thumbtack_hires	Integer	Scrapling / Thumbtack
nextdoor_recommendations	Integer	Scrapling / Nextdoor
ig_location_tag_count	Integer	Apify / Instagram Location Scraper — posts tagged at business location
ig_hashtag_mention_count	Integer	Apify / Instagram Hashtag Scraper — posts using business name hashtag
fb_checkin_count	Integer	Apify / Facebook Search Scraper — customer check-ins at business
fb_ugc_tag_count	Integer	Apify / Facebook Search Scraper — posts tagging business page
total_customer_ugc	Integer	Derived: sum of ig_location_tag + ig_hashtag_mention + fb_checkin + fb_ugc_tag

Outreach Record Entity (CRM Core)

Field	Type	Description
outreach_id	UUID	Primary key
business_id	FK → Business	Link to lead
status	Enum	Pipeline stage (see Section 6)
first_contact_date	Datetime	When AI agent initiated first call
contact_method	Enum	Voice / Email / SMS / In-Person
call_transcript	Text	Full transcript from voice agent
call_sentiment_score	Float (-1 to 1)	LLM-derived sentiment from transcript
call_disposition	Enum	Answered/Voicemail/No Answer/Wrong Number
meeting_scheduled	Boolean	Whether a meeting was booked
meeting_type	Enum	Virtual / In-Person
meeting_datetime	Datetime	Scheduled meeting time
follow_up_count	Integer	Number of follow-up attempts
last_contact_date	Datetime	Most recent outreach touch

assigned_to	String	Human team member for handoff
notes	Text	Free-form internal notes
proposal_amount	Float	Proposed monthly retainer if sent
contract_amount	Float	Actual signed amount if won
lost_reason	String	Reason for lost/disqualified status

3. Lead Scoring Model

3.1 Scoring Philosophy

The fundamental inversion: for these verticals, digital absence is the primary positive signal. Standard B2B lead scoring treats website visits, content downloads, and social engagement as indicators of purchase intent. That framework is irrelevant here. The businesses most likely to buy marketing services are the ones with the weakest digital footprints — they have the most to gain and the fewest existing vendor relationships blocking the sale.

The model produces two outputs: Acquisition Likelihood Score (0–100) predicting probability of closing, and Price Tier (1–3) predicting appropriate service level and price point. After voice outreach begins, a third signal — Call Sentiment Score — feeds back into the composite to adjust priority in real time.

3.2 Acquisition Likelihood Score

Digital Deficit Score (40% of total)

Measures the gap between current digital presence and minimum viable marketing infrastructure. Scored 0–100 where higher = greater deficit = greater need.

Signal	Condition	Points
No website	has_website = false	+30
Website exists but poor quality	website_quality_score < 40	+20
No SSL certificate	has_ssl = false	+8
No Google Business Profile	has_google_business_profile = false	+15
GBP incomplete	gbp_completeness_score < 0.5	+10
Zero Google reviews	google_review_count = 0	+10
Low review count	google_review_count < 10	+5
No social media presence	has_facebook_page = false AND has_instagram = false	+12
Dormant social media	fb_last_post_days_ago > 90	+8
Not running any paid ads	no Google Ads AND no Meta Ads	+7

Business Viability Score (35% of total) — REVISED WEIGHTS

Validates that the business is operational, stable, and capable of paying for services. The v2 model deprioritizes license status as a gating mechanism. Many legitimate businesses in these niches operate without or between license renewals — the presence of sustained customer reviews and years of continuous operation are stronger viability signals than bureaucratic compliance status.

Key change from v1: An expired or absent license no longer automatically disqualifies a lead. It reduces the score, but reviews, ratings, and operational longevity dominate.

Signal	Condition	Points	Rationale
In operation 3+ years	incorporation_date > 3 years ago	+20	Longevity is the strongest viability signal
In operation 7+ years	incorporation_date > 7 years ago	+10 (additional)	Survivor bias is informative here

Moderate review volume (10–50)	google_review_count between 10–50	+15	Sustained review flow = sustained foot traffic
High review volume (50+)	google_review_count > 50	+20	Strong revenue proxy
Rating above 4.0	google_avg_rating >= 4.0	+15	Quality signal; predicts retention
Rating 3.5–4.0	google_avg_rating >= 3.5	+8	Adequate quality
Positive review trajectory	review_velocity_30d > 0	+8	Business is growing
Customer UGC: moderate (10–50 tags)	total_customer_ugc between 10–50	+10	Location tags and check-ins are the hardest foot traffic signal to fake
Customer UGC: high (50+ tags)	total_customer_ugc > 50	+15	Strong organic customer engagement independent of business's own activity
Active license	license_status = active	+5	Supplementary compliance indicator
Nextdoor recommendations > 5	nextdoor_recommendations > 5	+5	Hyperlocal trust signal
Thumbtack hires > 10	thumbtack_hires > 10	+5	Direct transaction evidence
Multiple employees estimated	employee_count_est >= 3	+3	Revenue capacity indicator

A business with zero reviews across all platforms AND no incorporation/license record AND no Craigslist/Thumbtack activity is deprioritized (viability floor = 0). This replaces the prior hard gate on license status.

Competitive Pressure Score (25% of total)

Signal	Condition	Points
High competitor density	zip_niche_competitor_count > 5	+20
Very high density	zip_niche_competitor_count > 10	+10 (additional)
Competitors have stronger digital presence	business digital score < zip_niche_avg	+25
Competitors running paid ads	competitor_ads_active_count >= 2	+15
High-income zip code	zip_median_household_income > \$65,000	+10
High population density	zip_population_density > 10,000/sq mi	+10
Competitor avg rating higher	google_avg_rating < zip_niche_avg_rating	+10

Composite Score + Sentiment Feedback Loop

Base acquisition score = (digital_deficit * 0.40) + (viability * 0.35) + (competitive_pressure * 0.25)

After initial voice outreach, the Call Sentiment Score adjusts the composite:

- Positive sentiment (score > 0.3): acquisition_score * 1.15 (cap at 100)
- Neutral sentiment (-0.3 to 0.3): no adjustment
- Negative sentiment (score < -0.3): acquisition_score * 0.75
- No answer / voicemail after 2 attempts: acquisition_score * 0.90

This creates a dynamic scoring system where outreach data continuously refines lead priority rather than relying solely on pre-contact estimates.

3.3 Price Tier Estimation

During the bootstrap phase, pricing is negotiable and fit to each business's capability. The tier model below establishes reference ranges for outreach conversations, not hard floors. The goal is client acquisition and portfolio building — a \$150/month contract with a barbershop that becomes a referral source is worth more than a lost \$500 proposal. The voice agent and human follow-up should treat these tiers as starting points, not non-negotiable.

$$\text{est_monthly_revenue} = \text{niche_median_revenue} * (\text{zip_mhi} / \text{national_mhi}) * \text{review_volume_multiplier}$$

Tier	Reference Range	Conditions	Services Included
Tier 1	\$150–\$500	est_revenue < \$15K OR employees < 3 OR competitive_pressure < 30	GBP optimization, basic social media (2–3 posts/week), review solicitation
Tier 2	\$400–\$1,200	est_revenue \$15K–\$50K AND competitive_pressure 30–65 AND digital_deficit > 50	Tier 1 + website build, basic SEO, review management, Google Ads setup
Tier 3	\$900–\$2,500	est_revenue > \$50K OR competitive_pressure > 65 OR employees > 8	Tier 2 + paid ad management, content creation, advanced SEO, monthly reporting

Overlapping ranges are intentional. A Tier 2 business that balks at \$800 can be offered \$400 with a reduced scope. Record the actual contract amount in the CRM for model recalibration — the gap between predicted tier and accepted price is itself a training signal for future pricing accuracy. As the portfolio matures past the bootstrap phase, tighten the ranges and reduce flexibility.

4. System Architecture

4.1 Technology Stack

Component	Technology	Rationale
Orchestration	Python 3.12+	Ecosystem maturity for scraping and ML
Web scraping (static/API)	Scrapling	Adaptive element tracking survives layout changes; StealthyFetcher bypasses Cloudflare; anti-bot evasion out of the box; replaces Scrapy + Playwright
Web scraping (dynamic/JS)	Scrapling PlayWrightFetcher	Headless browser for DFPR, IL SOS form-based lookups
Platform scraping (Nextdoor, CL, Thumbtack, Angi)	Scrapling StealthyFetcher	Custom scrapers for non-Meta platforms; zero per-item cost; adaptive element tracking handles layout changes
Meta platform scraping (IG, FB, Threads)	Apify Actors (API)	Meta's anti-bot measures make custom scrapers cost-prohibitive to maintain; Apify handles proxy rotation and fingerprinting
LLM (local inference)	vLLM	High-throughput, memory-efficient serving via PagedAttention; OpenAI-compatible API; runs local models for entity resolution and extraction
LLM (complex tasks)	Claude API (Sonnet)	Outreach brief generation, GBP completeness assessment, nuanced entity resolution, sentiment analysis — tasks requiring superior reasoning
Voice outreach agent	Retell AI	Sub-700ms interruption handling, built-in sentiment scoring, CRM webhook integration, appointment scheduling, \$0.07–0.14/min all-in
Database	PostgreSQL 16 + PostGIS	Relational model fits entity structure; geo queries for zip code radius mode
Task queue	Celery + Redis	Async scraping jobs; rate-limit management
API layer	FastAPI	Internal API for leads, outreach triggers, CRM operations
CRM / Dashboard	React + Tailwind	Lightweight internal CRM with pipeline board, lead detail views, call logs
Containerization	Docker Compose	Reproducible local deployment; single docker-compose up

4.2 LLM Task Routing

Tasks are routed between vLLM (local) and Claude API based on complexity and cost sensitivity:

Task	Router	Model	Rationale
Entity resolution (name matching)	vLLM	Llama 3.1 8B or Qwen 2.5 7B	High volume, low complexity; local inference keeps costs near zero
Data normalization (address, phone)	vLLM	Llama 3.1 8B	Structured extraction with predictable patterns

Unstructured website extraction (email, owner)	vLLM	Llama 3.1 8B	Pattern extraction from HTML; volume-sensitive
GBP completeness assessment	Claude API	Sonnet	Requires nuanced evaluation of profile quality across multiple dimensions
Outreach brief generation	Claude API	Sonnet	Requires contextual reasoning about competitive dynamics and persuasion angles
Call transcript sentiment analysis	Claude API	Sonnet	Emotional nuance, sarcasm detection, intent classification
Revenue estimation refinement	vLLM	Llama 3.1 8B	Structured calculation with niche-specific heuristics

4.3 Nextdoor Cookie Refresh Automation

Nextdoor requires authenticated sessions to access business search results. Session cookies expire on a variable schedule (typically 7–14 days). Manual refresh is unsustainable. The system implements automated cookie rotation:

1. A headless Scraping PlaywrightFetcher session authenticates to Nextdoor using stored credentials on a Celery-scheduled task (every 5 days, conservatively inside the expiry window).
2. On successful authentication, the new session cookies are extracted and stored in Redis with a TTL matching the refresh interval.
3. All Nextdoor scraping tasks pull the current cookie set from Redis before initiating requests.
4. If authentication fails (CAPTCHA, account lock, credential change), the system logs the failure, alerts the operator, and marks Nextdoor enrichment as temporarily unavailable. Leads continue scoring without Nextdoor data — it degrades gracefully since Nextdoor recommendations carry only 5 points in the viability score.

A dedicated Nextdoor account (not a personal account) should be created for this purpose to isolate risk.

5. Voice Outreach System

5.1 Architecture

The voice outreach system uses Retell AI as the voice agent platform, integrated with the LeadForge pipeline via webhooks. The agent initiates outbound calls to scored leads, conducts a structured but conversational pitch, and attempts to schedule a virtual or in-person meeting.

Why Retell over alternatives

- Sub-700ms interruption handling — critical for cold calling skeptical small business owners who will test the agent with interruptions and objections.
- Built-in sentiment scoring per call — eliminates need for separate post-call analysis pipeline.
- Native appointment scheduling — can check availability and book directly during the call.
- Webhook-based CRM integration — call outcome, transcript, and sentiment push directly to LeadForge database.
- All-in pricing (\$0.07–0.14/min) — no hidden LLM/TTS/STT layered costs.

Fallback: Vapi (open-source, \$0.05/min + component costs) if full self-hosting becomes necessary.

Backend logic should be decoupled from the voice provider via webhook handler.

5.2 Call Flow

1. Agent initiates outbound call to lead's primary phone number.
2. If voicemail: leave a 15-second message with callback number and brief value proposition. Mark disposition as Voicemail. Schedule retry in 48 hours.
3. If answered: identify self as calling from [agency name], reference a specific observation about their business (generated from outreach brief). Example: "I noticed your barbershop has great reviews on Google but doesn't have a website — your competitors in the area do, and they're showing up in searches that could be sending customers your way."
4. Gauge interest. If positive or neutral, propose a 15-minute virtual call or in-person visit to discuss specifics.
5. If meeting agreed: collect preferred time, confirm virtual (Zoom/Google Meet) or in-person, and confirm contact info.
6. If declined: ask permission to follow up in 30 days. Mark disposition accordingly.
7. Post-call: transcript and sentiment score are pushed to LeadForge via webhook. Acquisition score is adjusted per sentiment feedback loop (Section 3.2).

5.3 Sentiment Analysis Integration

Every completed call produces a transcript that is analyzed by Claude Sonnet for:

- Overall sentiment (-1 to 1): hostile, dismissive, neutral, curious, interested, enthusiastic
- Objection classification: price concern, already have agency, not interested in marketing, bad timing, distrust of AI/cold calls
- Purchase intent signal: asked about pricing, asked about services, asked for examples, mentioned competitor
- Follow-up recommendation: immediate (hot lead), scheduled (warm), deprioritize (cold), disqualify (hostile/wrong contact)

This data feeds directly into the CRM record and adjusts the lead's composite acquisition score dynamically.

5.4 Compliance

- TCPA: Manual-equivalent dialing (one call at a time, no predictive dialer). Calls are to business lines, not personal mobile numbers, which reduces TCPA exposure. If the number is identified as a personal mobile, the system flags for human review before calling.
- CAN-SPAM: If email outreach is added later, it must include physical address, sender identification, and functional unsubscribe.
- Illinois law: No additional state-specific cold outreach restrictions apply to B2B solicitation.
- FTC Telemarketing Sales Rule: B2B calls are largely exempt, but the agent must identify itself honestly if asked whether it is AI.

6. CRM System

6.1 Design Philosophy

LeadForge's CRM is not Salesforce. It is purpose-built for a single pipeline: cold outreach to local small businesses for marketing services. It needs to track exactly one deal type through exactly one sales process with a small team. Anything beyond that is overhead. The CRM is a React + Tailwind frontend backed by FastAPI endpoints and the same PostgreSQL database as the scraping pipeline.

6.2 Pipeline Stages

Stage	Definition	Trigger
Scored	Lead has been scraped, enriched, and scored. No outreach yet.	Pipeline completion
Queued	Lead approved for outreach and queued for voice agent.	Operator approval or auto-queue for Tier A leads
Contacted	Voice agent has attempted first call.	Call initiated by Retell
Voicemail	Message left, awaiting callback or retry.	Retell disposition = voicemail
Engaged	Business owner answered and had a conversation.	Retell disposition = answered, sentiment > -0.3
Meeting Scheduled	Virtual or in-person meeting booked.	Agent confirmed meeting time
Proposal Sent	Pricing/scope proposal delivered after meeting.	Manual status update
Negotiating	Prospect responded to proposal with questions or counter.	Manual status update
Won	Contract signed, client acquired.	Manual status update
Lost	Business declined after engagement.	Manual or auto after 3 failed follow-ups
Disqualified	Invalid contact, closed business, or fundamentally wrong fit.	Manual or auto-detection
Nurture	Not ready now but open to future contact. Re-enter pipeline in 90 days.	Manual status update

6.3 CRM Feature Set

Pipeline Board View

Kanban-style board showing leads across pipeline stages. Cards display: business name, niche, zip code, acquisition score, price tier, days in current stage, and last contact date. Drag-and-drop between stages with automatic timestamp logging.

Lead Detail View

- Business profile: all scraped/enriched data, digital presence summary, competitive context
- Score breakdown: digital deficit, viability, competitive pressure sub-scores with contributing signals visible
- Outreach timeline: chronological log of all contacts — calls, voicemails, emails, meetings
- Call recordings + transcripts: linked from Retell, playable inline
- Sentiment history: graph of sentiment scores across all calls

- Notes: free-form notes field for human team members
- Outreach brief: LLM-generated pitch angle, regenerable on demand

Filters and Search

- Filter by: zip code, niche, pipeline stage, acquisition score range, price tier, date range, assigned team member
- Sort by: acquisition score, days in stage, last contact date, sentiment score
- Search: business name, owner name, phone, address

Reporting

- Pipeline conversion rates: stage-to-stage conversion with average time in each stage
- Score calibration: actual outcomes (Won/Lost) correlated with predicted acquisition scores
- Outreach efficiency: calls made, answer rate, meeting rate, close rate
- Revenue: total pipeline value, won value, average contract size by niche and tier

7. Geographic Targeting

7.1 Zip Code Input Modes

- Single zip code: Scrape and score one zip for all target verticals or a specified subset.
- Zip code list: Comma-separated or file-uploaded list of specific zip codes.
- Radius mode: Given a center point (address or zip), select all Chicago zip codes within N miles.

7.2 Recommended Initial Targeting

South Shore (60649), Auburn Gresham (60620), Chatham (60619), Humboldt Park (60647), Back of the Yards (60609), Pilsen (60608), Little Village (60623), and Englewood (60621). These areas have high small business density, lower digital maturity on average, and lower likelihood of existing agency relationships. Validate against actual Chicago Data Portal license counts before committing outreach resources.

8. Development Roadmap

Phase 1: Data Pipeline MVP (Weeks 1–4)

- Scraping setup with StealthyFetcher and PlaywrightFetcher for DFPR/SOS automation.
- Chicago Data Portal Socrata API integration (business licenses by zip + category).
- Google Places API enrichment.
- Basic digital deficit scoring (website, GBP, reviews).
- PostgreSQL + PostGIS data model. CSV export.
- CLI: `python leadforge.py --zip 60619 --niche barbershops`

Phase 2: Full Scoring + Platform Scrapers (Weeks 5–8)

- DFPR and IL SOS headless scraping via Scrapling.
- Custom Scrapling scrapers for Nextdoor (with automated cookie refresh), Craigslist, Thumbtack, and Angi.
- Apify integration for Meta platforms only: Instagram, Facebook, Threads, Meta Ad Library.
- Yelp Fusion, PageSpeed, WHOIS enrichment.
- Complete three-factor scoring model.
- Competitive context computation per zip-niche pair.
- Census data integration.
- vLLM deployment for entity resolution and extraction tasks.

Phase 3: LLM Integration + Voice Outreach (Weeks 9–12)

- Claude API integration for outreach briefs and GBP assessment.
- vLLM local inference for entity resolution, normalization, extraction.
- Retell AI integration: agent configuration, call flow scripting, webhook handler.
- Sentiment analysis pipeline (Claude Sonnet on call transcripts).
- Sentiment feedback loop into acquisition score.

Phase 4: CRM + Dashboard (Weeks 13–16)

- FastAPI endpoints for full CRUD on leads, outreach records, pipeline operations.
- React + Tailwind CRM: pipeline board, lead detail, filters, search.
- Call recording/transcript integration from Retell.
- Reporting views: conversion rates, score calibration, revenue tracking.
- Automated 90-day recalibration trigger: Celery scheduled task that fires at day 90 post-deployment and every 90 days thereafter. It pulls all Won/Lost outcomes, computes correlation between sub-score weights and actual close rates, generates a recalibration report with proposed weight adjustments, and queues it for operator review. If the operator approves, new weights are applied to all uncontacted leads. The trigger also flags if sample size is insufficient (< 50 Won + Lost outcomes) and recommends extending the cycle.

9. Success Metrics

Metric	Target	Measurement
Lead accuracy	> 85% valid current contact info	Spot-check 50 leads/batch

Score-to-close correlation	Tier A closes at 2x+ Tier C rate	Compare close rates by tier
Price tier accuracy	> 70% closed deals in predicted range (post-bootstrap)	Actual contract vs. predicted tier; bootstrap-phase deviations tracked separately
Voice agent answer rate	> 35% of calls answered	Retell call disposition data
Meeting scheduling rate	> 15% of answered calls → meeting	Retell appointment data
Outreach → close rate	> 5% from first contact to Won	Pipeline conversion tracking
Pipeline velocity	500 leads scraped+scored per batch < 2 hours	End-to-end pipeline timing
Client acquisition cost	< \$50 in time + API costs per Won client	Total pipeline cost / Won count
Sentiment accuracy	> 80% agreement with human review on 50-call sample	Human audit of sentiment classifications

10. Open Questions and Risks

1. Retell AI voice quality under load. Test with 50+ concurrent calls before committing to production outreach. If latency or quality degrades, Vapi is the self-hosted fallback.
2. Small business owner receptivity to AI callers. If answer rates are significantly below 35%, consider human-initiated first touch with AI handling scheduling and follow-up only.
3. Apify cost for Meta platforms specifically. With Craigslist, Thumbtack, Angi, and Nextdoor moved to custom Scraping scrapers, the remaining Apify spend is limited to Instagram, Facebook, and Threads enrichment. At projected volumes this should stay under \$80/month, but monitor and evaluate whether Meta enrichment data justifies even that cost after 90 days of outcome data.
4. Entity resolution accuracy across 8+ data sources. This is the highest-risk technical challenge. Validate LLM fuzzy matching against a 200-record manual audit before trusting at scale.
5. Nextdoor cookie refresh automation must be monitored for account-level risk. Nextdoor may flag or suspend accounts used for automated access. Use a dedicated account, rotate user-agent strings, and keep request volume below detection thresholds. If Nextdoor locks the account, the system degrades gracefully — leads continue scoring without the 5-point Nextdoor recommendation signal.
6. Bootstrap pricing flexibility introduces model noise. During the bootstrap phase, actual contract amounts will deviate significantly from predicted tiers because pricing is negotiated to the business's capability. The recalibration system must account for this by tracking both the predicted tier and the negotiated amount, and distinguishing between "priced down to close" and "tier prediction was wrong." After the bootstrap phase, tighten ranges and re-evaluate which deviations were strategic versus predictive errors.
7. Automated 90-day recalibration trigger is built into Phase 4. The system fires a Celery scheduled task at day 90 and every 90 days thereafter, correlating sub-score weights with actual Won/Lost outcomes and generating a recalibration report for operator review. If sample size is insufficient (< 50 terminal outcomes), the system extends the cycle and alerts the operator.
8. TCPA gray area with AI voice agents calling mobile numbers listed as business contacts. Legal review has been approved as a prerequisite before scaling outreach. Conservative approach during bootstrap: flag mobile numbers for human-initiated first touch. Reassess after legal counsel provides guidance.